

Project 1

ArtBench

Generative Modeling

Generative AI | 2025/2026 | 2nd Semester

João Correia · Penousal Machado · Pedro Louro · Sancho Simões

Deadline: 12 April 2026

What Is This Project About?

Train generative models on artwork images and produce new plausible samples that match the data distribution.

The Task

- Learn to generate images of paintings
- Multiple artistic styles (10 classes)
- Balance fidelity, diversity, and style

The Challenge

- Compare different model families
- Rigorous quantitative evaluation
- Statistical reporting across seeds

This is NOT classification — there is no single right answer. Quality is measured by distribution-level metrics.

Groups of 2 students · Individual grading

Dataset: ArtBench-10

A curated dataset of artworks designed to benchmark generative models in the visual arts domain.

10

Artistic Styles

50K

Training Images

32×32

RGB Resolution

Development Phase

- Use the provided 20% subset
- Faster iteration on architectures
- Explore hyperparameters and strategies

Final Evaluation

- Retrain best model on full dataset
- 50,000 training samples
- Final metrics computed here

Required Model Families

You must implement, train, and compare at least one model from each of these three families.

Autoencoders

At least one VAE-type model

Strengths:

- Stable training
- Smooth latent space

Weakness:

- Blurry outputs

GANs

At least one DCGAN model

Strengths:

- Sharp samples
- Fast generation

Weakness:

- Unstable training

Diffusion

At least one diffusion model

Strengths:

- High quality
- Good diversity

Weakness:

- Slow inference

Additional methods beyond these three are encouraged and may earn extra credit.

Evaluation Protocol

- 1 Generate 5,000 samples per model
- 2 Sample 5,000 real images from test split
- 3 Compute FID and KID
- 4 Repeat 10× with different random seeds
- 5 Report mean $\pm$ standard deviation

FID — Fréchet Inception Distance

- Full 5,000 generated vs 5,000 real
- Lower = better quality + diversity

KID — Kernel Inception Distance

- 50 subsets of size 100
- Report mean $\pm$ std across subsets

Min 10 seeds per config · Same preprocessing and eval code across all models · Include visual sample grids

Your Pipeline

Build a complete generative pipeline from data loading to evaluation.

Data
Loading

Pre-
processing

Model
Training

Sampling

Evaluation
(FID / KID)

Report

Development Strategy

1. Start with the 20% subset for fast iteration
2. Compare all three model families
3. Select best approach
4. Retrain on full 50K dataset
5. Final evaluation with 10 seeds

Starting Resources

- Starter codebase provided
- ArtBench paper (arXiv:2206.11404)
- HuggingFace diffusers tutorial
- PyTorch GAN Zoo (Facebook)
- Reference packs for each family

Report & Submission

Report Structure

- Introduction
- Problem description
- Methodology and approach
- Experimental setup
- Analysis of results
- Conclusion

Format & Rules

- Springer LNCS format (LaTeX)
- Max 12 pages (excl. references)
- Portuguese or English
- Submit via Infoforestudante
- Overleaf template available

What We Look For

- Clear description of adopted approach
- Model architectures and training details
- Reproducible protocol (hyperparams, seeds)
- Quantitative evaluation with proper metrics

AI Tool Policy

- You may use AI tools (code assistants, LLMs)
- All AI use must be explicitly acknowledged
- You are fully responsible for correctness

Summary & Key Dates

Deadline: 12 April 2026

Submit via Infoforestudante



Train at least 3 model families: VAE, GAN, Diffusion



Develop on 20% subset, final eval on full dataset



Compute FID and KID with 10 random seeds



Write a 12-page report in Springer LNCS format



Acknowledge all sources including AI tools



Public defence — individual grading

Good luck!